

A Journey Through Exponential Utility Maximization under Multivariate Fake Stationary Affine Volterra Models.

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Multivariate fake stationary affine Volterra processes with convolutive kernel

Fix $T > 0$, $d \in \mathbb{N}$, under the given complete filtered probability space $(\Omega, \mathcal{F}, \mathbb{F} = \{\mathcal{F}_t\}_{0 \leq t \leq T}, \mathbb{P})$, let $V = (V^1, \dots, V^d)^\top$ be the following $\mathbb{R}_+^d$ -valued scaled Volterra square-root process driven by an d -dimensional process $W = (W^1, \dots, W^d)^\top$:

$$V_t = \varphi(t) V_0 + \int_0^t K(t-s)(\mu(s) + DV_s) ds + \int_0^t K(t-s) \nu \varsigma(s) \sqrt{\text{diag}(V_s)} dW_s, \quad V_0 \perp\!\!\!\perp W. \quad (1)$$

Here, $K = \text{diag}(K_1, \dots, K_d)$ is diagonal with scalar kernels $K_i \in L^2([0, T], \mathbb{R})$, $\varphi = \text{diag}(\varphi^1, \dots, \varphi^d)$, $\nu = \text{diag}(\nu_1, \dots, \nu_d)$, $\varsigma = \text{diag}(\varsigma^1, \dots, \varsigma^d)$ with ς^i a (locally) bounded Borel function and $D := -\text{diag}(\lambda_1, \dots, \lambda_d) \in \mathbb{R}^{d \times d}$. $\mu : \mathbb{R}_+ \rightarrow \mathbb{R}^d$.

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Well-Posedness: Unique-in-law positive continuous weak solution (as a scaling limit of a sequence of time-modulated Hawkes processes) see e.g., [Gnabeyeu, Pagès and Rosenbaum, 2026]

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Well-Posedness: **Unique-in-law positive** continuous weak solution (as a **scaling limit** of a sequence of **time-modulated Hawkes processes**) see e.g., [Gnabeyeu, Pagès and Rosenbaum, 2026]

Definition (Fake Stationarity)

The process $(V_t)_{t \geq 0}$ with diffusion coefficient $\sigma(V)$, starting from $V_0 \in L^2(\mathbb{P})$, exhibit a **fake stationary regime of type I** if:

$$\forall t \geq 0, \quad \mathbb{E} V_t = \mathbb{E} V_0 =: m_0, \quad \text{Var}(V_t) = \text{Var}(V_0) =: v_0 \quad \text{and} \quad \mathbb{E} \sigma^2(V_t) = \bar{\sigma}_0^2. \quad (2)$$

Proposition (Fake stationary Volterra process.)

Let $(V_t)_{t \geq 0}$ be a solution to the scaled Volterra equation in its form (1) starting from any random variable $V_0 \in L^2(\Omega, \mathcal{F}, \mathbb{P})$. Then, a necessary and sufficient condition for the relations (2) to be satisfied is that for $i = 1, \dots, d$

$$\forall t \geq 0, \quad \varphi^i(t) = 1 - D_{ii} \int_0^t K_i(t-s) \left(\frac{\mu^i(s)}{D_{ii} m_0^i} - 1 \right) ds. \quad (3)$$

and each couple $(v_0^i, \varsigma^i(t))$, where $v_0^i = \text{Var}(V_0^i)$ must satisfy the functional equation: ($f_{D_{ii}}$ is the resolvent of $D_{ii} K_i$)

$$(E_{D_{ii}, c_i}): \forall t \geq 0, \quad c_i D_{ii}^2 \left(1 - (\varphi^i(t) - (f_{D_{ii}} * \varphi^i)_t)^2 \right) = (f_{D_{ii}}^2 * \varsigma^i{}^2)(t) \text{ where } c_i = \frac{v_0^i}{\nu_i^2 m_0^i} \text{ i.e. } \varsigma^i := \varsigma_{D_{ii}, c_i}^i. \quad (4)$$

Definition (Stabilizer)

We will call the **stabilizer (or corrector)** of the scaled stochastic Volterra equation (1) the (locally) bounded Borel function $\varsigma = \text{diag}(\varsigma^1, \dots, \varsigma^d)$ where ς^i is a solution (if any) to the **functional equation** (E_{D_{ii}, c_i}) in (4) for $i = 1, \dots, d$.

Example (Computing the Stabilizer $\varsigma_{\alpha_i, D_{ii}, c_i}$ for the α_i -fractional kernels $K_{\alpha_i}(t) = \frac{u^{\alpha_i-1}}{\Gamma(\alpha_i)} \mathbf{1}_{\mathbb{R}}(t)$)

Within the setting $\mu^i = D_{ii}m_0^i \Rightarrow \varphi^i(t) = \varphi^i(0) = 1$ for all $t \geq 0$ and $i = 1, \dots, d$, the functional equation (E_{D_{ii}, c_i}) reads :

$$(E_{D_{ii}, c_i}): \quad \forall t \geq 0, \quad c_i D_{ii}^2 \left(1 - \left(1 - \int_0^t f_{D_{ii}}(s) ds \right)^2 \right) = (f_{D_{ii}}^2 * \varsigma^{i2})(t). \quad (5)$$

Example (Computing the Stabilizer $\varsigma_{\alpha_i, D_{ii}, c_i}$ for the α_i -fractional kernels $K_{\alpha_i}(t) = \frac{t^{\alpha_i-1}}{\Gamma(\alpha_i)} \mathbf{1}_{\mathbb{R}}(t)$)

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Regular Variation (Tauberian theorem) on Laplace transforms of (E_{D_{ii}, c_i}) suggests to search $\varsigma^2(t)$ as an expansion of the form (Exponential Power Series Ansatz): Set $a_k = \frac{1}{\Gamma(\alpha_i k + 1)}$, $b_k = \frac{1}{\Gamma(\alpha_i(k+1))}$, $k \geq 0$.

$$\varsigma_{\alpha_i, D_{ii}, c_i}^2(t) = c_i D_{ii}^{2 - \frac{1}{\alpha_i}} \varsigma_{\alpha_i}^2(D_{ii}^{\frac{1}{\alpha_i}} t) \quad \text{where} \quad \varsigma_{\alpha_i}^2(t) := 2 t^{1-\alpha_i} \sum_{k \geq 0} (-1)^k c_k^i t^{\alpha_i k}. \quad (6)$$

with

$$c_0^i = \frac{\Gamma(\alpha_i)^2}{\Gamma(2\alpha_i - 1)\Gamma(2 - \alpha_i)}, \quad \text{and for every } k \geq 1, \quad c_k \text{ is defined inductively by:}$$

$$c_k^i = \frac{\Gamma(\alpha_i)^2 B(\alpha_i(k+1), 2(1-\alpha_i))}{\Gamma(2(1-\alpha_i)) \Gamma(2\alpha_i - 1)} \left[(a * b)_k - \alpha_i(k+1) \sum_{\ell=1}^k B(\alpha_i(\ell+2) - 1, \alpha_i(k-\ell-1) + 2) (b^{*2})_{\ell} c_{k-\ell} \right].$$

where for two sequences of real numbers $(u_k)_{k \geq 0}$ and $(v_k)_{k \geq 0}$, the Cauchy product is defined as $(u * v)_k = \sum_{\ell=0}^k u_{\ell} v_{k-\ell}$

Example: Computing the Stabilizer $\varsigma_{\alpha_i, D_{ii}, c_i}$ for the α_i -fractional kernels $K_{\alpha_i}(t) = \frac{t^{\alpha_i-1}}{\Gamma(\alpha_i)} \mathbf{1}_{\mathbb{R}}(t)$:

Proposition (Existence and Properties of the function $\varsigma_{\alpha_i, D_{ii}, c_i}^2$ for $\alpha_i \in (\frac{1}{2}, 1)$ [Pagès, 2024])

The *convergence radius* of the fractional power series (6) that defines $\varsigma_{\alpha_i, D_{ii}, c_i}$ is *infinite* and $\varsigma_{\alpha_i, D_{ii}, c_i}$ is positive on $[0, +\infty]$ so that $\varsigma_{\alpha_i, D_{ii}, c_i}$ is *well-defined*: The stabilizer $\varsigma_{\alpha_i, D_{ii}, c_i}^2$ exists as a non-negative function, such that:

$$\textcircled{1} \quad \varsigma_{\alpha_i, D_{ii}, c_i}(0) = 0 \quad \text{and} \quad \lim_{t \rightarrow +\infty} \varsigma_{\alpha_i, D_{ii}, c_i}(t) = \frac{\sqrt{c_i} D_{ii}}{\|f_{\alpha_i, D_{ii}}\|_{L^2(\text{Leb}_1)}}.$$

Example: Computing the Stabilizer $\varsigma_{\alpha_i, D_{ii}, c_i}$ for the α_i -fractional kernels $K_{\alpha_i}(t) = \frac{u^{\alpha_i-1}}{\Gamma(\alpha_i)} \mathbf{1}_{\mathbb{R}}(t)$:

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Numerical Illustrations: To simulate the volterra process with fractional kernel, we use the *K-integrated discrete time Euler* scheme on the time grid $t_k = t_k^n = \frac{kT}{n}$, $k = 0, \dots, n$, namely for $i = 1, 2$, $\bar{V}_0^{i,n} = V^{i,0}$ and for every $k = 1, \dots, n$,

$$\bar{V}_{t_k}^{i,n} = V^{i,0} + \underbrace{\sum_{\ell=1}^k (\mu_0^i - D_{ii} \bar{V}_{t_{\ell-1}}^{i,n}) \int_{t_{\ell-1}}^{t_{\ell}} K_{\alpha_i}(t_k - s) ds}_{(a)} + \underbrace{\sum_{\ell=1}^k \nu_i \varsigma_{\alpha_i, D_{ii}, c_i}(t_{\ell}) \sqrt{\bar{V}_{t_{\ell-1}}^{i,n}} \int_{t_{\ell-1}}^{t_{\ell}} K_{\alpha_i}(t_k - s) dW_s}_{(b)} \quad (7)$$

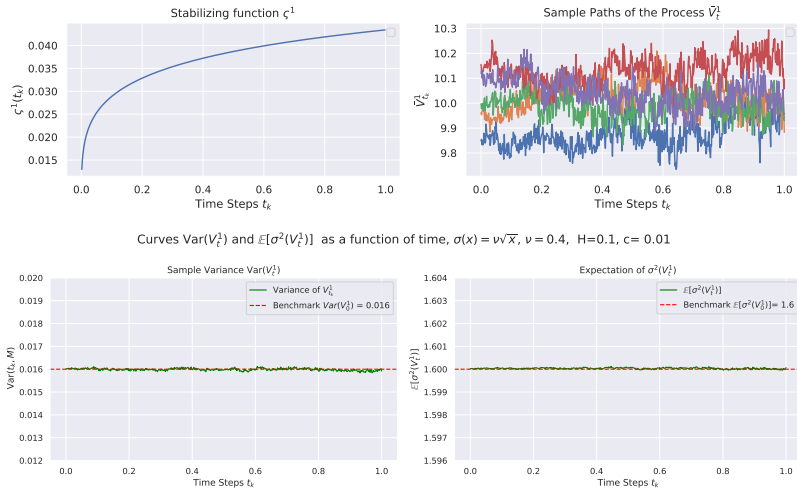


Figure: Stabilizer, sample paths, empirical variance, and mean squared volatility of V^1 on $[0, 1]$ ($H = 0.1$, $c_1 = 0.01$, $n = 600$).

Aim and Motivation:

- Maximize an investor's expected utility from terminal wealth with respect to a given utility function (U).
- One of the classical financial economic approaches to understand how volatility affects optimal investment decisions.

The research in applied optimal control with Volterra diffusion models has gained an increasing attention:

- Rough Heston (1D) with power utility: [Bäuerle and Desmettre, 2020]
- Volterra Heston (1D) with power and exponential utilities: [Han and Wong, 2020]
- Multidimensional Volterra Heston models with power utility: [Aichinger and Desmettre, 2021], based on a Verification argument using calculus of convolutions and resolvents.

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Challenges and Limitations:

- The 1D **Martingale distortion transformation** is no longer available (except in highly degenerate correlation settings).
- Extending the analysis to the multidimensional setting under a non-degenerate correlation structure.

Our contribution: [Gnabeyeu et al., 2026, Gnabeyeu, 2026]

- We solve the exponential utility maximization problem in multivariate fake stationary affine Volterra–Heston models.
- Solution of the problem by BSDEs and Martingale optimality principle, an approach initiated by [Hu et al., 2005]
- We recover the existing results from the literature as special cases.

Consider a financial market on $[0, T]$ with a non-risky asset satisfying an ODE

$$dS_t^0 = S_t^0 r(t) dt \quad (8)$$

and d risky assets (stock or index) whose return vector process $(S_t)_{t \geq 0} = (S_t^1, \dots, S_t^d)_{t \geq 0}$ is defined via the dynamics given by the vector-stochastic differential equation (SDE):

$$dS_t = \text{diag}(S_t) [r(t) \mathbf{1}_d dt + \sigma(V_t)(dB_t + \lambda_t dt)], \quad (9)$$

driven by a d -dimensional Brownian motion B and $r : \mathbb{R}_+ \rightarrow \mathbb{R}$, a time-dependent deterministic short risk-free rate.

- $\sigma(V.)$: $\mathbb{R}^{d \times d}$ -valued continuous stochastic volatility process whose dynamics is driven by (1),
- λ : $\mathbb{R}^d$ -valued stochastic market price of risk.

We assume the correlation structure of W with B is given by

$$W = \Sigma B + \sqrt{I - \Sigma^\top \Sigma} B^\perp, \quad \text{with} \quad \Sigma := \text{diag}(\rho_1, \dots, \rho_d) \quad (10)$$

for some $(\rho_1, \dots, \rho_d) \in [-1, 1]^d$, and $B^\perp = (B^{\perp,1}, \dots, B^{\perp,d})^\top$ is an d -dimensional Brownian motion independent of B

Exponential Utility Maximization: The Framework

• Let π_t denote the vector of the amounts invested in the risky assets S at time t in a self-financing strategy. Wealth X^π of the portfolio we seek to optimize is given by

$$dX_t^\pi = X_t^\pi (r(t) + \pi_t^\top \sigma(V_t) \lambda_t) dt + \pi_t^\top \sigma(V_t) dB_t. \quad (11)$$

Letting $\alpha_t := \sigma^\top(V_t) \pi_t$ be the investment strategy, the wealth X^α reads:

$$dX_t^\alpha = (r(t)X_t^\alpha + \alpha_t^\top \lambda_t) dt + \alpha_t^\top dB_t, \quad t \geq 0, \quad X_0^\alpha = x_0 \in \mathbb{R}. \quad (12)$$

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- The set of all admissible investment strategies is denoted as $\mathcal{A}$ and is naturally defined by:

$$\mathcal{A} = \left\{ (\alpha_t)_{t \in [0, T]} \in L_{\mathbb{F}}^{2, loc}([0, T], \mathbb{R}^d) \text{ such that (12) has a solution } X_t^\alpha \text{ for which the family } \left\{ \exp \left[-\gamma \int_\tau^T r(u) du \right] X_\tau^\alpha : \tau \text{ stopping time valued in } [0, T] \right\} \text{ is uniformly integrable.} \right\} \quad (13)$$

where given the complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ satisfying the usual conditions, we denote by

$$L_{\mathbb{F}}^p([0, T], \mathbb{R}^d) = \left\{ Y : \Omega \times [0, T] \mapsto \mathbb{R}^d, \mathbb{F} - \text{prog. measurable s.t. } \mathbb{E} \left[\int_0^T |Y_s|^p ds \right] < \infty \right\}$$
$$S_{\mathbb{F}}^p([0, T], \mathbb{R}^d) = \left\{ Y : \Omega \times [0, T] \mapsto \mathbb{R}^d, \mathbb{F} - \text{prog. measurable s.t. } \mathbb{E} \left[\sup_{0 \leq t \leq T} |Y_t|^p \right] < \infty \right\}.$$

Exponential Utility Maximization: BSDEs and Martingale Optimality Principle

Find the value function $\mathcal{V}^\xi(x_0, V_0)$ for the CARA utility function such that

$$\mathcal{V}^\xi(x_0, V_0) = \sup_{\alpha(\cdot) \in \mathcal{A}} \mathbb{E}_{x_0, V_0} [U(X_T^\alpha - \xi)], \quad U(x) := -\frac{1}{\gamma} \exp(-\gamma x), \quad \gamma > 0 \quad (14)$$

where ξ is a $\mathcal{F}_T$ -measure random terminal condition and $\gamma > 0$ the risk aversion coefficient of the investor.

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Similar to [Hu et al., 2005], we consider the family of stochastic processes $(J^\alpha)^\alpha$ defined for every $t \in [0, T]$ by

$$J_t^\alpha := -\frac{1}{\gamma} \exp(-\gamma x_0 e^{\int_0^T r(s) ds}) \exp\left(-\gamma \int_0^t e^{\int_s^T r(u) du} \left(\alpha_s^\top dB_s + \alpha_s^\top \lambda_s ds\right) + \gamma Y_t\right) \quad (15)$$

where the pair (Y, Λ) satisfies a Backward SDEs under $\mathbb{P}$ with a driver $f : [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ of the form:

$$\begin{cases} dY_t &= -f(t, Y_t, \Lambda_t) dt + \Lambda_t^\top dW_t, \\ Y_T &= \xi =: \int_0^T \sum_{i=1}^d \xi_i V_s^i ds. \end{cases} \quad (16)$$

$\Rightarrow$ We are bound to choose a function f for which J_t^α is a supermartingale for all $\alpha \in \mathcal{A}$ and there exists a $\alpha^* \in \mathcal{A}$ such that $J_t^{\alpha^*}$ is a martingale.

Proposition

For any $T > 0$, an optimal investment strategy $(\alpha_t^*)_{t \in [0, T]}$ for the exponential utility maximization problem (14) is given by

$$\alpha_t^* = e^{-\int_t^T r(s)ds} \left(\frac{\lambda_t}{\gamma} + \Sigma \Lambda_t \right), \quad 0 \leq t \leq T. \quad (17)$$

where the pair (Y, Λ) is a solution (if any) to the quadratic BSDE (18)

$$\begin{cases} dY_t &= \left(\frac{1}{2\gamma} |\lambda_t + \gamma \Sigma \Lambda_t|^2 - \frac{\gamma}{2} |\Lambda_t|^2 \right) dt + \Lambda_t^\top dW_t. \\ Y_T &= \int_0^T \sum_{i=1}^d \xi_i V_s^i ds. \end{cases} \quad (18)$$

Sketch of Proof: We verify that $\{J_t^\alpha\}_{t \in [0, T]}$, $\alpha \in \mathcal{A}$ satisfies the Martingale optimality principle in the sense that.

- ① $J_T^\alpha = U(X_T^\alpha - \xi)$ for all $\alpha \in \mathcal{A}$;
- ② J_0^α is a constant, independent of $\alpha \in \mathcal{A}$;
- ③ J_t^α is a supermartingale for all $\alpha \in \mathcal{A}$, for $\alpha^* \in \mathcal{A}$, J^{α^*} is a martingale (with some additional assumptions).

In this case,

$$\mathbb{E}_{x_0, v_0} [U(X_T^\alpha - \xi)] = \mathbb{E}_{x_0, v_0} [J_T^\alpha] \leq J_0^\alpha = J_0^{\alpha^*} = \mathbb{E}_{x_0, v_0} [J_T^{\alpha^*}] = \mathbb{E}_{x_0, v_0} [U(X_T^{\alpha^*})] = \mathcal{V}^\xi(x_0, v_0). \quad (19)$$

- There is **no general existence results** covering such **quadratic BSDE** (unbounded random coefficients)
- We assume that the risk premium λ and the stock volatility σ in (9) are **linear in $\sqrt{\text{diag}(V)}$** i.e., are in the form¹:

$$\sigma(V_t) = \sqrt{\text{diag}(V_t)} \quad \text{and} \quad \lambda_t = \left(\theta_i \sqrt{V_t^i} \right)_{1 \leq i \leq d}, \quad 0 \leq t \leq T. \quad (20)$$

for some constant $\theta_i \geq 0$.

⇒ This specification is in the **spirit of Heston-type model** (**Fake stationary Volterra Heston volatility**).

¹the BSDE admits a solution characterized by Riccati equations, say now **Riccati BSDE**

Proposition

Assume that there exists a solution $\psi \in C([0, T], \mathbb{R}^d)$ to the inhomogeneous Riccati-Volterra equation:

$$\psi^i(t) = \int_0^t K_i(t-s) \left(\xi_i - \frac{\theta_i^2}{2\gamma} + F_i(T-s, \psi(s)) \right) ds, \quad i = 1, \dots, d, \quad (21)$$

$$F_i(s, \psi) = -\theta_i \rho_i \nu_i \varsigma^i(s) \psi^i + (D^\top \psi)_i + \gamma \frac{\nu_i^2}{2} (1 - \rho_i^2) (\varsigma^i(s) \psi^i)^2. \quad (22)$$

Let Λ be defined as

$$\Lambda_t^i := \nu_i \varsigma^i(t) \psi^i(T-t) \sqrt{V_t^i}, \quad i = 1, \dots, d, \quad 0 \leq t \leq T. \quad (23)$$

Then, for some $p > 1$, (Y, Λ) is a $\mathbb{S}_{\mathbb{F}}^p([0, T], \mathbb{R}) \times L_{\mathbb{F}}^2([0, T], \mathbb{R}^d)$ -valued solution to the Riccati BSDE (18) and Y reads:

$$Y_t = \sum_{i=1}^d \int_0^t \xi_i V_s^i ds + \int_t^T \sum_{i=1}^d \left(\xi_i - \frac{\theta_i^2}{2\gamma} + F_i(s, \psi(T-s)) \right) g_t^i(s) ds, \quad (24)$$

where the $\mathbb{R}^d$ -valued process $(g_t(s))_{t \leq s}$, $g = (g^1, \dots, g^d)^\top$ is the adjusted forward variance process defined by:

$$g_t(s) = \mathbb{E}^{\mathbb{P}} \left[V_s - \int_t^s K(s-u) DV_u du \mid \mathcal{F}_t \right], \quad \mathbb{P}\text{- a.s., for a.e. } s > t. \quad (25)$$

Remark

Under the condition that $\xi_i - \frac{\theta_i^2}{2\gamma} < 0 \ \forall i = 1, \dots, d$, standard arguments provides the *existence of a unique global continuous solution* $\psi \in C([0, T], \mathbb{R}^d)$ to the inhomogeneous Ricatti–Volterra Equation (21)–(22) such that $\psi^i(t) < 0$ for every $t > 0$.

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This leads to our main result:

Theorem

Let ψ be the unique, continuous solution of the inhomogeneous Ricatti-Volterra equation (21)–(22) on the time interval $[0, T]$. Then, an optimal investment strategy $(\alpha_t^*)_{t \in [0, T]}$ for the portfolio exponential utility maximization problem (14) is given by

$$\alpha_t^* = \left(\frac{1}{\gamma} e^{-\int_t^T r(s)ds} (\theta_i + \gamma \rho_i \nu_i \zeta^i(t) \psi^i(T-t)) \sqrt{V_t^i} \right)_{1 \leq i \leq d}, \quad 0 \leq t \leq T. \quad (26)$$

Moreover α^* is admissible. The value function defined in (14) can be written as

$$\mathcal{V}^\xi(x_0, V_0) = -\frac{1}{\gamma} \exp \left(-\gamma e^{\int_0^T r(s)ds} x_0 + \gamma \int_0^T \sum_{i=1}^d \left(\xi_i - \frac{\theta_i^2}{2\gamma} + F_i(s, \psi(T-s)) \right) g_0^i(s) ds \right). \quad (27)$$

Numerical Experiments: Impact of Roughness and Risk Aversion

2D fake stationary rough Heston volatility: ($d=2$) and two different levels of roughness.

- Recall that $\alpha_t^* = \sigma(V_t)^\top \pi_t^*$ i.e. the optimal strategies $(\alpha_t^*)_{t \in [0, T]}$ are stochastic processes,
- We consider the evolution of the associated deterministic map $t \mapsto \pi_t^*$,

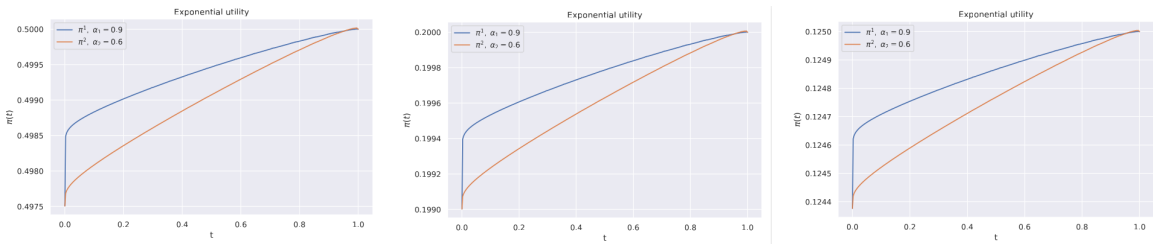


Figure: Optimal strategy for horizon $T = 1$ and risk aversion levels $\gamma = 0.2$, $\gamma = 0.5$, and $\gamma = 0.8$ (from left to right). Increasing risk aversion leads to a more conservative allocation.

Our illustrations reveal that:

- the optimal allocation is **moderated** as the volatility paths become rougher,
- the **lower** the investor's risk aversion, the higher the optimal allocation.

Numerical Experiments: Impact on the investment horizon

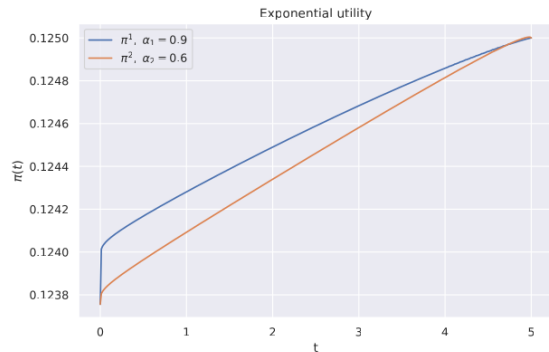
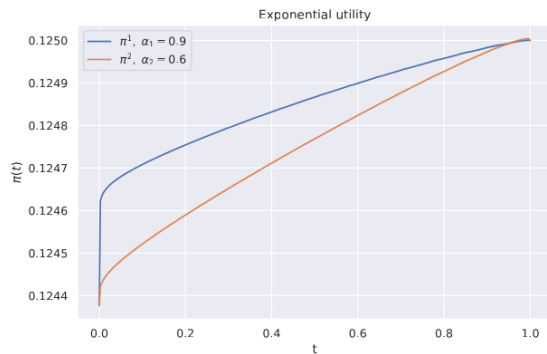


Figure: Effect of the time horizon T on the optimal strategy for risk aversion $\gamma = 0.8$ ($T = 1$ (left) and $T = 5$ (right)). The form of the control is consistent across different time scales.

Regardless of the investment horizon, the stabilizer induces the **same functional form for the optimal strategies, reflecting a time-scale invariance property**: the functional form of the optimal control is preserved across different investment horizons.



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Thanks For Your Attention!

Questions ?